

Lecture 14

Theory of General Linear System IV: Dimension and Rank

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Part 0 Review and Outline

2 mins

Basis

Definition 13.1 (basis)

Suppose V is a linear space over $\mathbb{R}$.

Suppose $\mathcal{U} \triangleq \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\} \subseteq V$. We say $\mathcal{U}$ is a basis if

- (i) $\mathcal{U}$ is linearly independent;
- (ii) $\text{span}(\mathcal{U}) = V$.

In words, a basis of V is linearly independent & spanning set of V .

Eg1 $\{\mathbf{u}\}$ is a basis of $\text{span}\{\mathbf{u}\}$, if $\mathbf{u} \neq \mathbf{0}$.

Eg2 $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is a basis of $\mathbb{R}^n$.
(called “standard basis”)

Major Result: Solution of $Ax=b$ and Solution of $Ax=0$

The following result holds not only for the example, but general system.

Claim: If $\mathbf{x}_p$ is a solution of $Ax=b$, then any solution of $Ax=b$ can be written as $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$, where $\mathbf{x}_n \in N(A)$.

General Proof:

Suppose $\mathbf{x}$ is a solution of $A\mathbf{x} = b$,

Then $A\mathbf{x} = b$ (1)

We also have $A\mathbf{x}_p = b$ (2).

Denote $\mathbf{x}_n = \mathbf{x} - \mathbf{x}_p$. From (1)-(2), we get

$$\mathbf{0} = b - b = (A\mathbf{x} - A\mathbf{x}_p) = A(\mathbf{x} - \mathbf{x}_p) = A\mathbf{x}_n.$$

Thus $\mathbf{x}_n \in N(A)$.

Today's Lecture: Outline

Strang's book: Sec 3.3, 3.4

Today ... Dimension, rank

1. Dimension
2. Rank

After this lecture, you should be able to

1. Explain why the dimension of a linear space is a well-defined
2. Compute the dimension and rank of simple spaces
3. Compute the number of solutions for different (r, m, n)

Part I Dimension

Strang's book Sec. 3.4

Part I Dimension

Strang's book Sec. 3.4

Part I.1 Size of Bases

18 mins

Size of Bases

$\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is a basis of $\mathbb{R}^3$

$\{\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3\}$ is a basis of $\mathbb{R}^3$

$\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_3\}$ is a basis of $\mathbb{R}^3$

What do you observe? They seem to have the same size.

Vote

Q1: Size of the smallest basis of $\mathbb{R}^3$? **A) 3** **B) 2** **C) 1**

Q2: Size of the biggest basis of $\mathbb{R}^3$? **A) ∞** **B) 3** **C) >3**

Dimension = Size of Bases

Theorem 14.1 (bases have same size)

If $\{\mathbf{u}_1, \dots, \mathbf{u}_m\}$ and $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ are bases of a linear space V , then $m = n$.

First “big”
Supporting Theorem
in this course!

All bases have the same size!!

To prove it, we just need the following lemma:

If a basis with n elements exists, then any $(n+1)$ elements are dependent.

Lemma 14.1

Suppose $m > n$. If $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis of a linear space V , then any m elements $\mathbf{u}_1, \dots, \mathbf{u}_m$ of V are linearly dependent.

This section is mainly devoted to prove Lemma 14.1

Can skip in class. Students can read themselves.

Proof of Theorem 14.1 Using Lemma 14.1

Proof of Theorem 14.1:

Prove by contradiction.

Assume the contrary, that, $m \neq n$.

Case 1: $m > n$.

By Lemma 14.1,

$m > n$ & there's a basis with n elements, $\{v_1, v_2, \dots, v_n\}$

$\Rightarrow \underline{u_1, \dots, u_m}$ are linearly dependent.

$\Rightarrow \underline{\{u_1, \dots, u_m\}}$ is not a basis.

A contradiction to the assumption.

Case 2: $m < n$.

Similar contradiction.

Conclusion: The assumption $m \neq n$ does not hold, thus $m = n$. Q.E.D

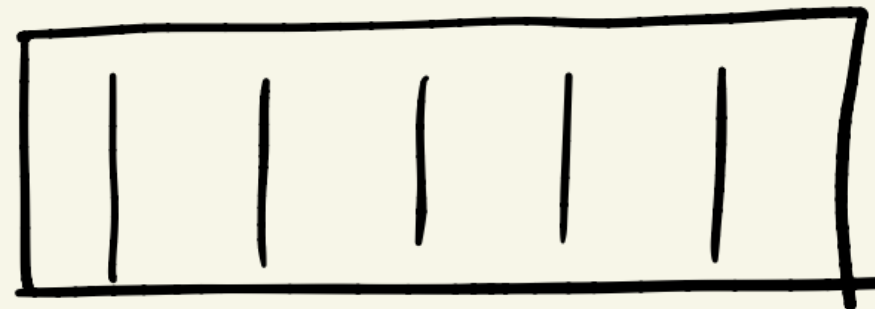
Lemma 14.1 in Euclidean Space

Lemma 14.1 (restated)

Suppose $m > n$. If $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a **basis** of a linear space V , then any m elements $\mathbf{u}_1, \dots, \mathbf{u}_m$ of V are linearly dependent.

Lemma 14.1E (Euclidean space version):

Any $(n + 1)$ vectors in $\mathbb{R}^n$ are linearly dependent.



Columns of a wide matrix
are linearly **DEPENDENT!!**

Exercise: Three 2-dim vectors are dependent

Exercise 1: Columns of $A = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 1 & 2 \end{bmatrix}$ are linearly dependent.

Want $\vec{x} \neq 0$, s.t. $x_1 \vec{a}_1 + x_2 \vec{a}_2 + x_3 \vec{a}_3 = 0 \Leftrightarrow Ax = 0$.

Equivalent to: $Ax = 0$ has **nonzero** solution.

Proof: Use GJ-E.

$$\begin{aligned} \left[\begin{array}{ccc|c} 2 & 4 & 6 & 0 \\ 1 & 1 & 2 & 0 \end{array} \right] &\xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 2 & 4 & 6 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 2 & 2 & 0 \end{array} \right] \xrightarrow{R_2/2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \\ &\xrightarrow{R_1 \rightarrow R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \end{aligned}$$

Correspond to $\begin{cases} x_1 + x_3 = 0 \\ x_2 + x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -x_3 \\ x_2 = -x_3 \end{cases}$

Any $(-t, -t, t)$ is a solution.

Thus, $Ax = 0$ has **∞ -many** solution

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What about these matrices?

$$\begin{bmatrix} 2 & 4 & 6 \\ 1 & 1 & 3 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} x & x & x & 0 \\ x & x & x & 0 \end{array} \right]$$

$$\begin{bmatrix} 2 & 3 & 6 \\ 1 & 2 & 5 \end{bmatrix}$$

↓ GJ-E

$$\left[\begin{array}{ccc|c} 1 & 0 & x & 0 \\ 0 & 1 & x & 0 \end{array} \right]$$

free var

$$\rightarrow \begin{cases} x_1 = 0 - 0x_3 \\ x_2 = 0 - 0x_3 \end{cases}$$

free

$$\begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \end{bmatrix}$$

or

$$\left[\begin{array}{ccc|c} 1 & x & x & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

free var

$$\rightarrow \begin{cases} x_1 = 0 + 0x_2 - 0x_3 \\ 0 = 0 \end{cases}$$

free

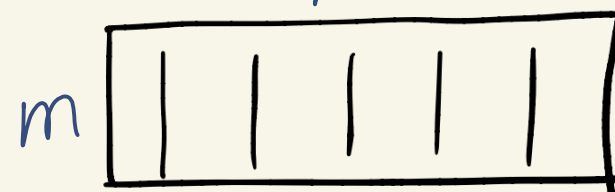
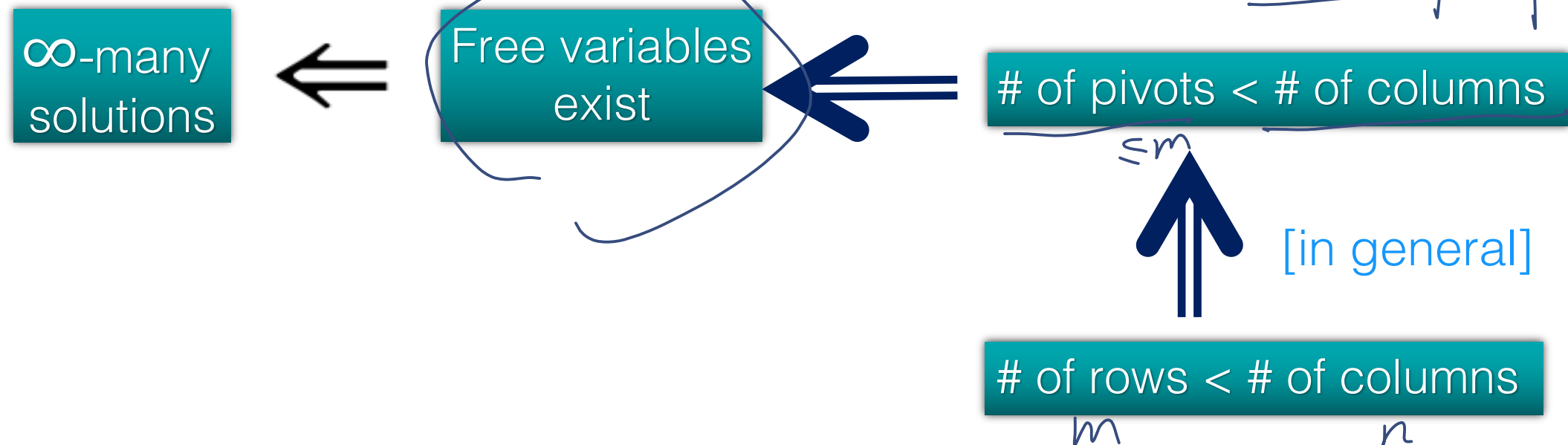
- 2 eq, 3 variable,

too much freedom ∞ -many solutions,

Proof Sketch of Lemma 14.1E

Lemma 14.1E says: Wide $Ax=0$ has a non-zero solution. *> n-m free variables*

Key Logic Chain of Proof:



Columns of a wide matrix
are linearly **DEPENDENT!!**

Part I Dimension

Strang's book Sec. 3.4

Part I.2 Definition of Dimension

15 mins

Dimension = Size of Bases

All bases have the same size!! We call it “dimension”!

Definition 14.1 (**dimension**)

Suppose V is a linear space.

If V has a basis $\mathcal{U}$ with n elements, then we say the dimension of V is n , denoted as $\dim(V)=n$, or V is n -dimensional.

Eg: $\dim(\mathbb{R}^n) = n$, since $\{\mathbf{e}_i, i = 1, \dots, n\}$ is the standard basis
 n elements

$\dim(\mathbb{R}^{2 \times 3}) = 6$, since $\{\mathbf{E}_{ij}, 1 \leq i \leq 2, 1 \leq j \leq 3\}$ is a basis.
 6 elements

Clarification: Dimension of Space & Matrix

We mentioned before.

$\begin{bmatrix} 2 & 3 & 1 \\ 1 & 4 & 5 \end{bmatrix}$ is a matrix of dimension 2×3 .

(don't recommend)

or a matrix of size 2×3 .

$\left\{ \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \mid a_{ij} \in \mathbb{R} \right\} \leadsto$ a **SPACE** of
dim 6.

Reading: Two Extreme Cases

Extreme Case 1:

Consider a linear space V , and suppose $\mathbf{0}$ is its zero element.

--Can be a zero vector, a zero matrix, or a zero polynomial, etc.

$U = \{\mathbf{0}\}$ is a linear space.

What is its dimension?

Answer: We say its dimension is 0 , since...

$\{\mathbf{0}\}$ is linearly dependent (exists $c=1$ s.t. $c \mathbf{0} = \mathbf{0}$).

Thus $\{\mathbf{0}\}$ is not a basis of $\{\mathbf{0}\}$.

Extreme Case 2:

If there is no finite set of elements that can span a linear space V , we say V is infinite dimensional.

Exercise

What is the dimension of the following linear spaces?

$$P_n = \left\{ \sum_{i=0}^n a_i x^i \mid a_i \in \mathbb{R} \right\} \quad \cancel{n} \quad n+1$$

a basis $\{1, x, x^2, x^3, \dots, x^n\}$
(n+1) functions.

eg

P_2 : basis

$\{1, x, x^2\}$.

3 functions.

$$C \left(\begin{array}{cc|c|c|c} 1 & 0 & a & 0 & c \\ 0 & 1 & b & 0 & d \\ 0 & 0 & 0 & 1 & e \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

Column
space

e_1, e_2, v_1, e_3, v_2

$\cancel{5}$

3

$= \text{span}\{e_1, e_2, e_3\}$.

$$C(A) = \text{span}\{e_1, e_2, e_3, v_1, v_2\}.$$

Want a basis of $C(A)$

Claim $\{e_1, e_2, e_3\}$ is a basis of $C(A)$

- indep: skip.

- span:

$$v_1 = a e_1 + b e_2 \in \text{span}\{e_1, e_2, e_3\}$$

$$v_2 = c e_1 + d e_2 + e e_3 \in \text{span}\{e_1, e_2, e_3\}.$$

Part II Dimension

Strang's book Sec. 3.4

Part II.3 Dimension, Span, Independence and Invertibility

5 mins

Quicker Way to Check Basis

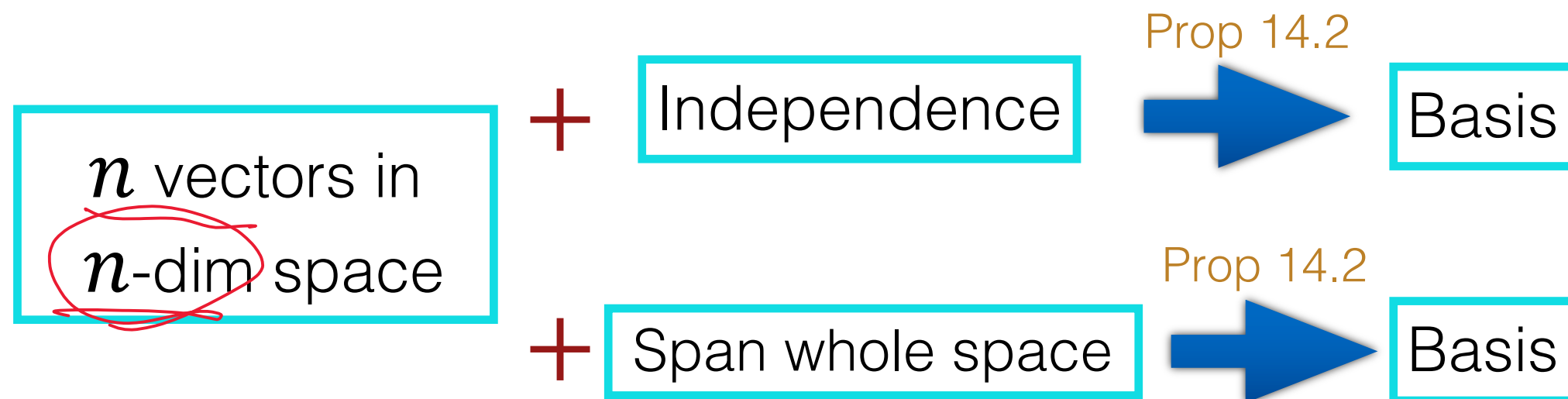
Proposition 14.2

[skip proof]

Consider a linear space V with $\dim(V) = n$.

- (a) If $\mathbf{u}_1, \dots, \mathbf{u}_n$ are linearly independent, then they form a basis of V .
- (b) If $\mathbf{u}_1, \dots, \mathbf{u}_n \in V$ can span V , then they form a basis of V .

In words: For n vectors, only need to check either “span” or “lin. indep.”, not both.



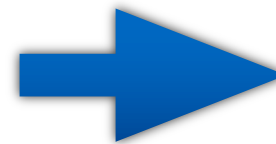
Quicker Way to Check Basis

E.g. $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ are linearly independent;
they are 2 vectors in $\mathbb{R}^2$; so they form a basis of $\mathbb{R}^2$.

2 vectors in
2-dim space $\mathbb{R}^2$

+

Independence



Basis

E.g. $v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$

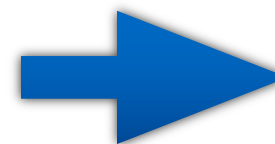
$v_1 = e_1, v_2 - 2v_1 = e_2, v_3 - 3v_1 = e_3$.
 $u = \alpha e_1 + \beta e_2 + \gamma e_3 = \alpha' v_1 + \beta' v_2 + \gamma' v_3$
can span $\mathbb{R}^3$;

they are 3 vectors in $\mathbb{R}^3$; so they form a basis of $\mathbb{R}^3$.

3 vectors in
3-dim space

+

Span whole space



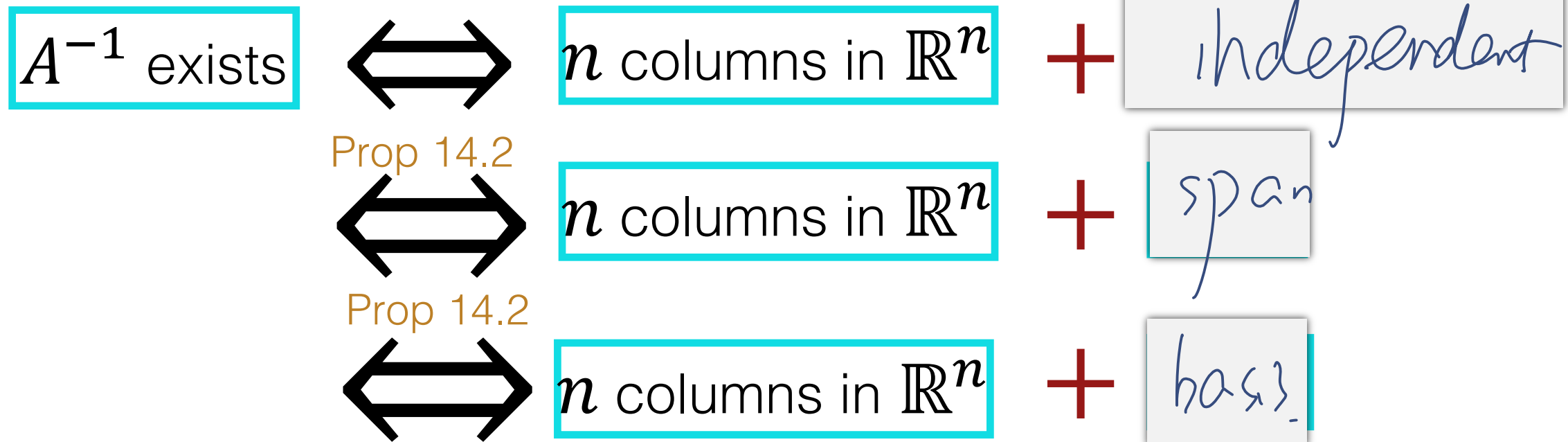
Basis

E.g. $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ are linearly independent, thus they form a basis of $\mathbb{R}^3$.

Two More Invertibility Condition

Recall: Lec 13: $n \times n$ matrix A is invertible iff n columns are linearly independent.

Lec 13's result



Corollary: $n \times n$ matrix A is invertible iff

- i) Its columns form a basis of $\mathbb{R}^n$.
- ii) Its columns span the whole space, i.e. $C(A) = \mathbb{R}^n$;
- iii) $Ax=b$ is solvable for any b .

(ii) $\iff$ (iii) proof:

By Lec 13, " $Ax=b$ has a solution" $\iff$ b is in $C(A)$
 thus " $Ax=b$ has a solution" for any $b \in \mathbb{R}^n$ $\iff$ any $b \in \mathbb{R}^n$ is in $C(A)$
 $\iff C(A) = \mathbb{R}^n$

Invertibility Conditions

Theorem 14.2a (Equivalent Conditions for Invertibility)

Let $A \in \mathbb{R}^{n \times n}$.

The following statements are equivalent:

1. A is invertible
2. The linear system $A\mathbf{x} = \mathbf{0}$ has a unique solution $\mathbf{x} = \mathbf{0}$
3. A is a product of elementary matrices
4. A has n pivots.
5. The columns of A are linearly independent;
6. The columns of A span $\mathbb{R}^n$.
7. The columns of A form a basis of $\mathbb{R}^n$;
8. $\dim(C(A)) = n$.
9. $A\mathbf{x} = \mathbf{b}$ is solvable for any $\mathbf{b}$

Equation solving

Linear independence

Span

Basis

Dimension

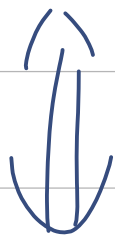
Equation solving

Most major concepts we learned so far are related to invertibility.

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For $n \times n$ A

$AX = 0$ has unique solution $X = 0$



$AX = b$ is solvable for any b .

Part II Rank

Strang's book Sec. 3.4

- Theorem: Row rank = Column rank = rank
- Number of solutions for different ranks

Part II Rank

Strang's book Sec. 3.4

Part II.1 Row Rank = Column Rank

15 mins

Row Space

Definition (row space)

Consider a matrix $A = \begin{bmatrix} \mathbf{a}_{(1)}^\top \\ \vdots \\ \mathbf{a}_{(m)}^\top \end{bmatrix} \in \mathbb{R}^{m \times n}$.

$\text{span}(\{\mathbf{a}_{(1)}, \dots, \mathbf{a}_{(m)}\}) \subseteq \mathbb{R}^n$ is called the row space of A , denoted as **Row(A)**.

A's row space is the span of A's row vectors.

Remark: $\text{Row}(A) = C(A^T)$.

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A's row space is the span of A's row vectors.

Remark: Row(A) = C(A^T).

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

E.g. Row(I_n) = $\mathbb{R}^n$.

E.g. Row $\left(\begin{bmatrix} 1 & 0 \\ 4 & 3 \\ 2 & 5 \end{bmatrix}\right) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix} \right\}.$

Row Rank, Column Rank and Rank

Definition (**row-rank and column rank**)

Row-rank of A is defined as $\dim(\text{Row}(A))$, denoted as $r_R(A)$.

Column-rank of A is defined as $\dim(\text{C}(A))$, denoted as $r_C(A)$.

Theorem 14.3

Row-rank, column-rank ~~and rank of a matrix~~ are the same, i.e.,

$$r_R(A) = r_C(A).$$

Equivalent expression: **$\dim(\text{Row}(A)) = \dim(\text{C}(A))$** .

Row Rank, Column Rank and Rank

Definition (row-rank and column rank)

Row-rank of A is defined as $\dim(\text{Row}(A))$, denoted as $r_R(A)$.

Column-rank of A is defined as $\dim(\text{C}(A))$, denoted as $r_C(A)$.

Theorem 14.3

Row-rank, column-rank ~~and rank of a matrix~~ are the same, i.e.,

$$r_R(A) = r_C(A).$$

Equivalent expression: $\dim(\text{Row}(A)) = \dim(\text{C}(A))$.
 $\leq m$ $\leq n$

Definition (rank) *FF*

The rank of a matrix A is $r_R(A)$, which is also $r_C(A)$.
Denoted as $\text{rank}(A)$.

Property: For $m \times n$ matrix, $\text{rank}(A) \leq \min\{m, n\}$.

Example 1 of Row-Rank = Column-Rank

Example 1:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 8 & 12 \end{bmatrix}$$

Rows:

$$a_{(1)} = [1, 2, 3]$$

$$a_{(2)} = [4, 8, 12] = 4 a_{(1)}$$

$$\text{So, dim(row space)} = 1$$

Row(A) is a line in $\mathbb{R}^3$.

row space
= $\text{span}\{\vec{a}_{(1)}, 4\vec{a}_{(1)}\}$

Columns:

$$a_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, a_2 = \begin{bmatrix} 2 \\ 8 \end{bmatrix} = 2a_1, a_3 = \begin{bmatrix} 3 \\ 12 \end{bmatrix} = 3a_1$$

$$\text{So, dim(column space)} = 1 \quad C(A) \text{ is a } \underline{\text{line}} \text{ in } \mathbb{R}^2.$$

$\text{span}\{\vec{a}_1, 2\vec{a}_1, 3\vec{a}_1\}$

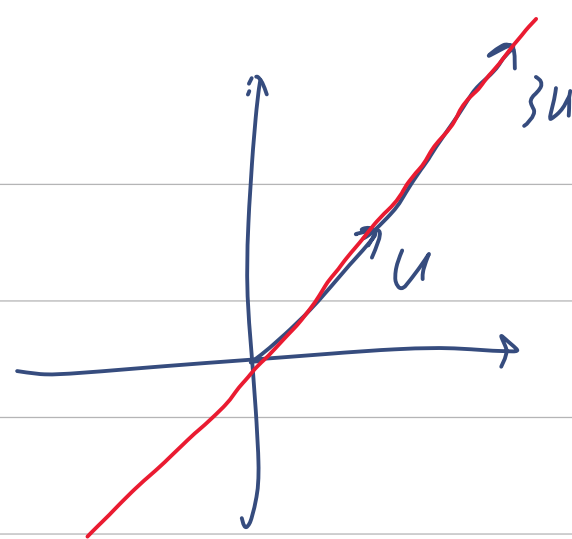
Is it a **coincidence** that rows are parallel & columns are parallel?

Thm 14.3 says: No.

Whenever rows are parallel, columns must be parallel too!

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$$\vec{u} \neq 0,$$



$$\dim(\text{span}\{\vec{u}, 3\vec{u}\}) = 1.$$

since $\{\vec{u}\}$ is a basis of $\text{span}\{\vec{u}, 3\vec{u}\}$.

$$\left(\begin{array}{l} - \text{span}\{\vec{u}\} = \text{span}\{\vec{u}, 3\vec{u}\}. \\ - \{\vec{u}\} \text{ is l.m. indep. } (\vec{u} \neq 0) \end{array} \right)$$

Example 2 of Row-Rank = Column-Rank

Example 2:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

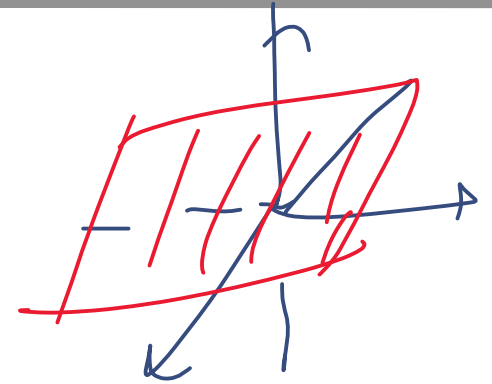
Rows:

$$a_{(1)} = [1, 2, 3]$$

$$a_{(2)} = [1, 1, 1]$$

$$\text{Dim}(\text{row space}) = 2$$

Row(A) is a plane in $\mathbb{R}^3$.

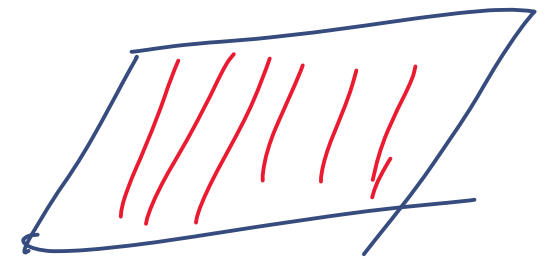


Columns:

$$a_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, a_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, a_3 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \in \mathbb{R}^2$$

$$\text{Dim}(\text{column space}) = 2$$

C(A) is a plane in $\mathbb{R}^2$, i.e., the entire $\mathbb{R}^2$



Even though they live in different ambient spaces, their dimensions are equal!

Rank and # of Pivots

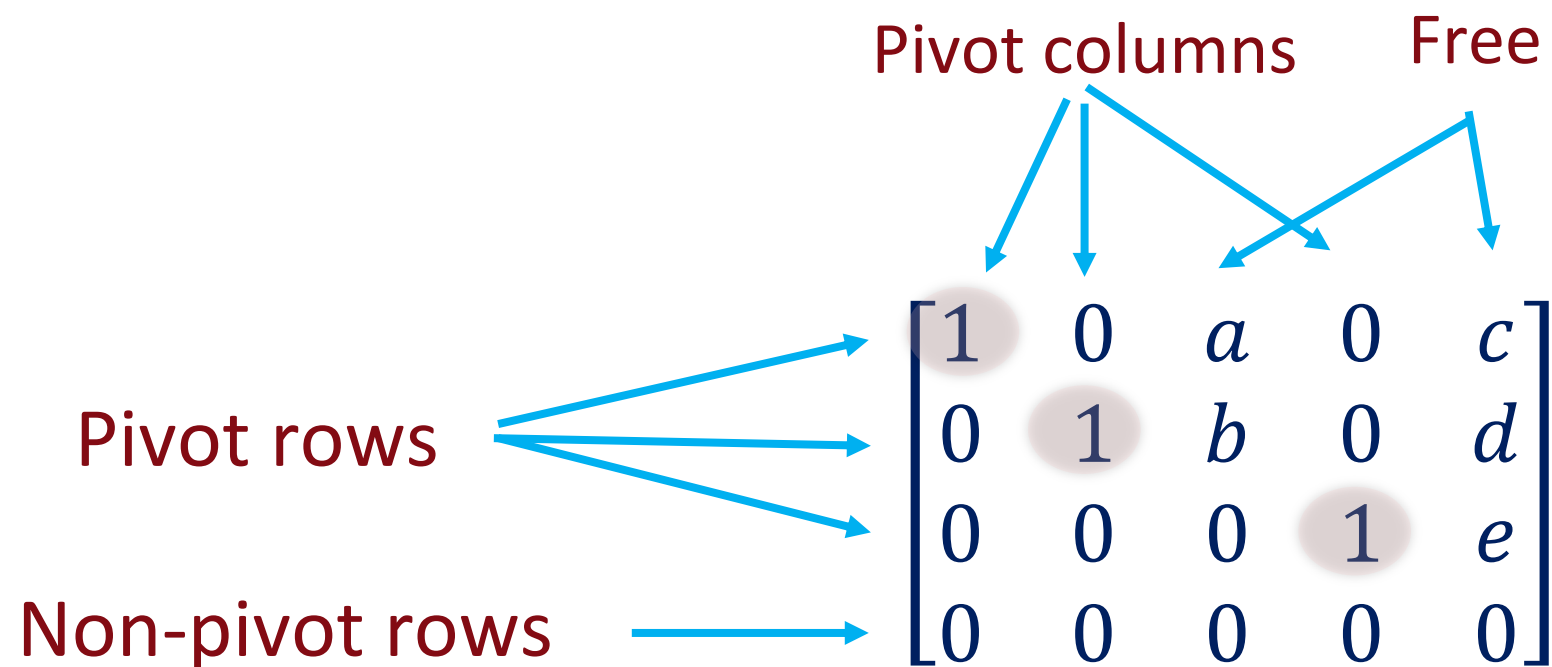
Proposition (rank = # of pivots)

Suppose when using Gauss-Jordan elimination to solve a linear system $Ax = 0$, there are ~~rank(A)~~ pivots.

Then $\text{rank}(A) = r$.

Skip the proof.

For RREF, this is relatively easy to see.



of pivots = 3

$\dim(C(A)) = 3$

Row-Rank = Column-Rank = # of pivots

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 3 & 3 \\ 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

(Handwritten notes: An arrow points from x^2 to the first row. Red wavy lines are drawn under the first, third, and fourth rows. A red oval encircles the second column.)

$$\text{Row}(A) = \text{Span} \left([1, 2, 0, 0]^T, [0, 0, 1, 0]^T, [0, 0, 0, 1]^T \right)$$

$$r_R(A) = 3.$$

$$\text{C}(A) = \text{Span} \left(\begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 1 \\ 0 \end{bmatrix} \right)$$

$$r_C(A) = 3.$$

After GJ-E, get RREF:

$$R = \begin{bmatrix} \boxed{1} & 2 & 0 & 0 \\ 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3 pivots

Thus, $r_R(A) = r_C(A) = 3 = \text{rank}(A)$.

【Reading】

Proof Sketch of Thm 14.3

Lemma 14.2

Suppose M is in an RREF. Then [skip proof here]

$$r_R(M) = r_C(M) = \text{Number of pivots.}$$

Claim: Elementary row operations do not change row space dimension, and do not change column space dimension. [skip proof here]

Proof of Theorem 14.3:

Suppose after GJ-E for matrix A , obtain RREF M .

Then by the above claim, we have and .

By Lemma 14.2, we obtain Done.

Part II Rank

Strang's book Sec. 3.4

Part II.2 Number of solutions v.s. Rank

(Optional; leave to next lecture if no time)

10 mins

Number of Solutions v.s. Rank

Assume the pivot columns appear at the beginning.

(if not, the analysis is still similar; omit here)

Under this assumption, the RREF is written as:

$$\begin{matrix} r \\ m-r \end{matrix} \left\{ \begin{bmatrix} I_r & F \\ \mathbf{0}_{(m-r) \times r} & \mathbf{0}_{(m-r) \times (n-r)} \end{bmatrix} \begin{bmatrix} x_P \\ x_F \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{b}}_{r \times 1} \\ \mathbf{c}_{(m-r) \times 1} \end{bmatrix} \right. \quad \text{Here, \# of pivots } r = \text{rank.}$$

Dependent on how large n, m, r , some of the blocks may disappear.

Four types	$\begin{bmatrix} I \end{bmatrix}$	$\begin{bmatrix} I & F \end{bmatrix}$	$\begin{bmatrix} I \\ 0 \end{bmatrix}$	$\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$
Their ranks	$r = m = n$	$r = m < n$	$r = n < m$	$r < m, \quad r < n$
# of solutions	1	∞	0 or 1	0 or ∞
# of solutions when $\mathbf{c}_{(m-r) \times 1} = \mathbf{0}$	1	∞	1	∞

more rigorously: N/A since C doesn't exist

When $\mathbf{c}_{(m-r) \times 1} = \mathbf{0}$, equations $0 = 0$ are redundant

Number of Solutions v.s. Rank

Case 2: $r=m < n$.

$$\begin{bmatrix} I & F \end{bmatrix}$$

$$\begin{bmatrix} \underline{I} & F \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}_p} \\ \mathbf{x}_f \end{bmatrix} = \hat{\mathbf{b}}$$

→ appear in Part I

$$\mathbf{x}_p + F\mathbf{x}_f = \hat{\mathbf{b}} \implies \mathbf{x}_p = \hat{\mathbf{b}} - F\mathbf{x}_f$$

Free variables ($\mathbf{x}_f$) can take any value in $\mathbb{R}^{n-r}$.

Conclusion: ∞ many solutions.

Case 3: $r=n < m$.

$$\begin{bmatrix} I \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} I \\ 0 \end{bmatrix} \mathbf{x}_p = \begin{bmatrix} \hat{\mathbf{b}} \\ \mathbf{c} \end{bmatrix}$$

$$\begin{cases} \mathbf{x}_p = \hat{\mathbf{b}}, \\ 0 = \mathbf{c}. \end{cases}$$

If $\mathbf{c} = 0$: exactly one solution.
If $\mathbf{c} \neq 0$: no solution.

Conclusion: 0 or 1 solution(s).

Number of Solutions v.s. Rank

Case 4: $r < m, r < n$.

$$\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_p \\ \mathbf{x}_f \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{b}} \\ \mathbf{c} \end{bmatrix}$$

$$\begin{cases} \mathbf{x}_p + F\mathbf{x}_f = \hat{\mathbf{b}}, \\ 0 = \mathbf{c}. \end{cases} \implies \begin{cases} \mathbf{x}_p = \hat{\mathbf{b}} - F\mathbf{x}_f \\ 0 = \mathbf{c}. \end{cases}$$

If $c = 0$: ∞ -many solutions.
If $c \neq 0$: no solution.

Conclusion: 0 or ∞ -many solution(s).

Concluding Part

Strang's book Sec. 3.4

How to Fill the Gap Between Courses and Homework & Exam?

First, figure out all logic & proofs of the lectures.

Second, work on more practice problems.

—Different from high school:

High school: Teachers (almost) provide you all needed problems.

University: You need to seek more practice problems yourself.
[Active learning]

—Where? Textbook practice problems.

—Too many? Pick a few that appear hard for you to work on.

Review Questions

How many **propositions** do we have today? [4]

How many **theorems** do we have today?

What are they?

Which is the most nontrivial one?

How many definitions do we have today?

What are they?



Summary Today (write Your Own)

One sentence summary:

Detailed summary:

Summary Today (of Instructor)

One sentence summary:

Basis, Dimension, Rank.

Detailed summary:

1. Basis and Dimension

—Dimension of linear space = size of any basis.

—Well defined due to Theorem 14.1:

Any basis has the same size (size = # of vectors in the basis).

2. Rank

Column/row-rank = dim of column/row space

Rank theorem: row-rank = column-rank = rank (= # of pivots)

3. Number of solutions

Depend on the shape $[I, F; 0, 0]$, which depend on n, m, r .

Appendix: Finishing Proof of Row Rank = Column Rank

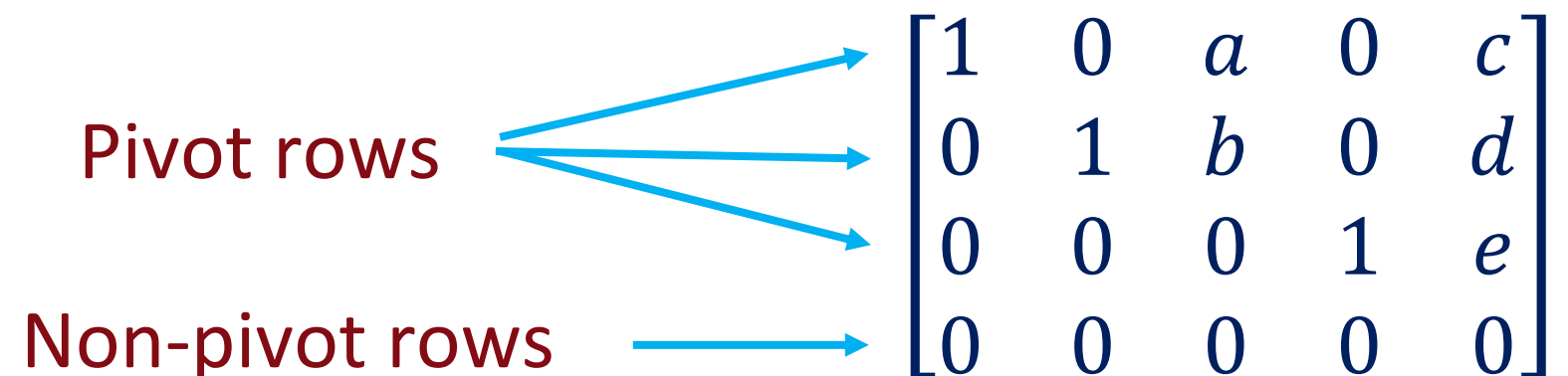
Prove:

Claim: Elementary row operations do not change row space dimension, and do not change column space dimension.

In RREF: Row Rank = # of Pivots

Pivot rows

Non-pivot rows

$$\begin{bmatrix} 1 & 0 & a & 0 & c \\ 0 & 1 & b & 0 & d \\ 0 & 0 & 0 & 1 & e \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$


First, span. We prove $\text{Span}\{\text{rows}\} = \text{span}\{\text{pivot rows}\}$.

Because Non-pivot rows are

Second, independence. We prove: Pivot rows are linearly independent.

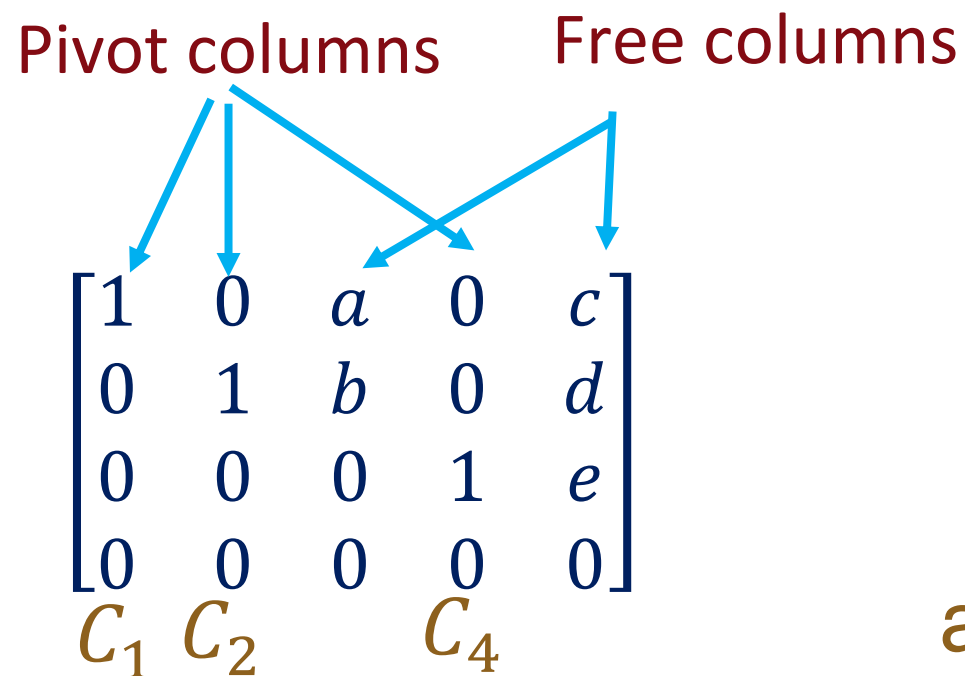
If $\alpha_1 R_1 + \alpha_2 R_2 + \alpha_3 R_3 = 0$, then 1st, , entry = 0, i.e., $\alpha_1 = \text{} = \text{} = 0$

Together, we get:

Pivot rows form a basis of row space.

$\Rightarrow \dim(\text{Row}(R)) = \# \text{ of pivot rows} = \# \text{ of pivots.}$

1000



$$\begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ 0 \end{bmatrix} = f_1 C_1 + \text{free column}$$

First, span. We prove: $\text{Span}\{\text{all columns}\} = \text{span}\{\text{pivot columns}\}$.
Only need to show: any free column is a LC of pivot columns.

Second, independence. We prove: Pivot columns are linearly independent.
If $\alpha_1 C_1 + \alpha_2 C_2 + \alpha_3 C_4 = 0$, then 1st, , entry = 0, i.e., $\alpha_1 =$ = 0

Together, we get:

Pivot columns form of .

$\Rightarrow \dim(C(R)) = \# \text{ of } \text{>} = \# \text{ of pivots.}$

Elementary Row Operation does Not Change LC Relation

Pivot columns

$$\begin{bmatrix} 1 & 0 & a & 0 & c \\ 0 & 1 & b & 0 & d \\ 0 & 0 & 0 & 1 & e \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} -a \\ -b \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Free columns

Consider the matrix $\hat{R}$ that becomes R after one step of elementary row operation.

$$\hat{R} = \begin{bmatrix} 1 & 0 & a & -\beta & c - \beta e \\ 0 & 1 & b & 0 & d \\ 0 & 0 & 0 & 1 & e \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\beta R_3 + R_1} \begin{bmatrix} 1 & 0 & a & 0 & c \\ 0 & 1 & b & 0 & d \\ 0 & 0 & 0 & 1 & e \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\Leftrightarrow (-c)\mathbf{v}_1 + (-d)\mathbf{v}_2 + (-e)\mathbf{v}_4 + \mathbf{v}_5 = 0.$$

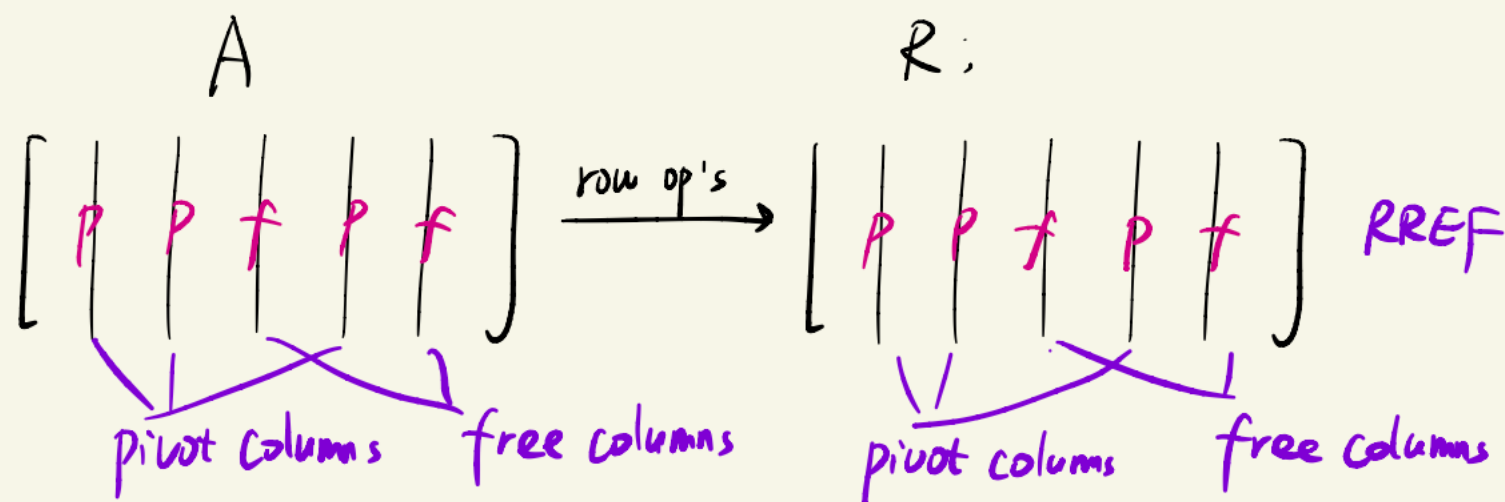
$$\text{Check: } (-c)\mathbf{v}_1 + (-d)\mathbf{v}_2 + (-e)\hat{\mathbf{v}}_4 + \hat{\mathbf{v}}_5 =$$

Claim 14.1: Elementary row operation does not change “linear combination” relation.

i.e. if $\alpha_1 \cdot (\text{Column } 1) + \cdots + \alpha_n \cdot (\text{Column } n) = 0$ holds for M , then it holds for the row-transformed matrix too.

Finishing Proof

to prove $\dim(C(A)) = \dim(C(R)) = r$.



r pivot cols
of A form
a basis of $C(A)$

$\Downarrow$
 $\dim(C(A)) = r.$

pivot columns of A
span free columns of A

pivot columns of A
are lin. independent

$\Leftarrow$ Claim 14.1

pivot columns of R
span free columns of R

pivot columns of R
are lin. independent

$\Leftarrow$ Claim 14.1